Nepal Engineering Council License Examination

Model Test 1

PANA ACADEMY

FM: 100 and PM: 50

1.In a series circuit with a fixed voltage, if the resistance is increased, how is the power dissipated across the load affected?

a) It increases proportionally

b) It decreases proportionally

c) It remains constant

d) It becomes zero

2

2. Which of the following statements is true regarding Kirchhoff's Voltage Law (KVL)?

a) The sum of the currents entering a junction is equal to the sum of the currents leaving the junction.

b) The sum of the voltage drops in a closed circuit is equal to the sum of the voltage rises.

c) The total voltage in a series circuit is equal to the sum of the individual voltages across each component.

d) Both b and c are true.

4

3.In an RLC circuit, what is the condition for resonance to occur?

a) The inductive reactance must equal the capacitive reactance.

b) The total impedance must be zero.

c) The resistance must be maximum.

d) The power factor must be equal to 1.

1

4. What is the main advantage of using Thevenin's theorem in circuit analysis?

a) It simplifies the calculation of power dissipation.

b) It allows for the calculation of complex waveforms.

c) It reduces the circuit to a simple equivalent circuit for easier analysis.

d) It eliminates the need for passive components.

3

5. In a circuit with a capacitor and resistor in series, if the frequency of the AC supply increases, what happens to the capacitive reactance (Xc)?

a) It increases

b) It decreases

c) It remains constant

d) It becomes zero

2

6. Which of the following configurations of a BJT is used for high-frequency applications due to its lower capacitance?

a) Common Base

b) Common Emitter

c) Common Collector

d) Emitter Follower

1

7.What is the binary equivalent of the decimal number 13?

A) 1010

B) 1110

C) 1101

D) 1011

3

8.How many select lines are required for an 8-to-1 multiplexer?

A) 2

B) 3

C) 4

D) 8

2

9.The main difference between combinational and sequential logic circuits is that sequential circuits:

A) Require a clock signal

B) Have no feedback

C) Depend only on current inputs

D) Are faster than combinational circuits

1

10. Which of the following is true about microprocessors?

a) It has an internal memory

b) It has interfacing circuits

c) It contains ALU, CU, and registers

d) All of the above

3

11.The continuous transfer may be interrupted by an external device by pulling down the signal

a) HRQ

b) DACK (active low)

c) DACK (active high)

d) HLDA

4

12.What is the purpose of an interrupt in a microprocessor system?

A) To initiate data transfer

B) To stop all processes

C) To execute a specific subroutine

D) To increase processing speed

3

13. Which of the following is a fundamental data type in C programming?

A) String

B) Array

C) Integer

D) Class

3

14. In C programming, what does a pointer store?

A) Integer value

B) Memory address

C) String value

D) Function definition

2

15. Which access specifier in C++ allows access to all member functions?

A) Public

B) Private

C) Protected

D) Friend

1

16. Which feature of object-oriented programming allows a function to be used in more than one way?

A) Encapsulation

B) Inheritance

C) Operator overloading

D) Class

3

17. Which keyword is used in C++ to define a virtual function?

A) Abstract

B) Public

C) Static

D) Virtual

4

18. Which feature of C++ allows a template to be used for generic programming?

A) Inheritance

B) Friend function

C) Class template

D) Inline function

3

19. Which unit controls the sequencing of operations in a CPU?

A) ALU

B) Control Unit

C) Memory Unit

D) Input Unit

2

20. Which of the following is an internal memory?

A) Hard Disk

B) Cache Memory

C) CD-ROM

D) USB Drive

2

21. Which of the following uses Direct Memory Access (DMA)?

A) CPU only

B) Peripheral devices

C) I/O Devices

D) Multiprocessor

3

22. What is an embedded system?

A) A general-purpose computer

B) A system designed for specific control functions

C) A high-performance server

D) A gaming console

2

23. In a real-time system, which scheduling algorithm is commonly used?

A) First-Come, First-Serve

B) Round Robin

C) Priority Scheduling

D) Rate Monotonic Scheduling

4

24.What does the std_logic type represent in VHDL?

A) A binary value (0 or 1)

B) A signal that can take on multiple values (U, X, 0, 1, Z, W, L, H)

C) A floating-point number

D) A fixed-point number

2

25. Which OSI layer is responsible for data encryption?

A) Network Layer

B) Data Link Layer

C) Physical Layer

D) Presentation Layer

4

26. Which multiple access protocol is used in Ethernet?

A) CSMA/CD

B) Token Ring

C) ALOHA

D) FDMA

1

27. Which protocol is used for dynamic routing?

A) RIP

B) FTP

C) SMTP

D) HTTP

1

28. Which port number is used by the HTTP protocol?

A) 21

B) 80

C) 25

D) 443

2

29. Which encryption algorithm is used in securing email communication?

A) AES

B) RSA

C) PGP

D) DES

3

30. What is the primary purpose of a firewall?

A) To protect data from unauthorized access

B) To store data

C) To speed up network performance

D) To allocate IP addresses

1

31. Which of the following is used to recognize patterns in a finite automaton?

A) Regular expression

B) Context-free grammar

C) Pushdown automaton

D) Turing machine

1

32.Which of the following is a characteristic of context-free languages?

A) They can be generated by finite automata

B) They can be parsed using a pushdown automaton

C) They can only be expressed using regular expressions

D) They cannot have nested structures

2

33. Which machine can simulate any other machine's computation?

A) DFA

B) Turing Machine

C) Pushdown Automaton

D) Mealy Machine

2

34. What is the basic unit for graphics in a raster display?

A) Line

B) Pixel

C) Shape

D) Vertex

2

35.When performing 2D transformations, which coordinate system is typically used?

A) Cartesian coordinates

B) Polar coordinates

C) Homogeneous coordinates

D) Spherical coordinates

3

36. Which projection method preserves parallel lines in 3D rendering?

A) Perspective Projection

B) Parallel Projection

C) Orthogonal Projection

D) Isometric Projection

2

37.Which data structure uses the Last In, First Out (LIFO) principle?

A) Stack

B) Queue

C) Linked List

D) Tree

1

38. Which sorting algorithm is based on the divide-and-conquer technique?

A) Selection Sort

B) Bubble Sort

C) Insertion Sort

D) Merge Sort

4

39. In a relational database, which key uniquely identifies a record in a table?

A) Foreign Key

B) Candidate Key

C) Primary Key

D) Composite Key

3

40. Which property of transactions ensures all operations are completed successfully or none at all?

A) Atomicity

B) Consistency

C) Isolation

D) Durability

1

41.What is the purpose of a process control block (PCB)?

A) To store the data of a process

B) To keep track of process status and control information

C) To execute system calls

D) To manage memory allocation

2

42. Which page replacement algorithm replaces the page that hasn't been used for the longest time?

A) FIFO

B) LFU

C) LRU

D) Random Replacement

3

43. Which software development model uses short iterative cycles?

A) Agile Model

B) Waterfall Model

C) V-Model

D) Spiral Model

1

44. Which document specifies the functional and non-functional requirements of a system?

A) Use Case Document

B) System Requirement Specification (SRS)

C) Design Document

D) Test Plan

2

45. Which design concept emphasizes dividing software into modules?

A) Modular Decomposition

B) Waterfall Design

C) Structural Design

D) Procedural Design

1

46. What type of testing is performed by developers during the coding phase?

A) System Testing

B) User Acceptance Testing

C) Integration Testing

D) Unit Testing

4

47. Which diagram represents the interactions between objects in an object-oriented system?

A) Class Diagram

B) Sequence Diagram

C) Collaboration Diagram

D) Use Case Diagram

3

48. Which OOP principle allows a derived class to inherit properties from a base class?

A) Encapsulation

B) Polymorphism

C) Inheritance

D) Abstraction

3

49.What is the main component of an intelligent agent's architecture?

A) Input and Output Devices

B) Sensors and Actuators

C) Algorithm and Data

D) Decision Table

2

50.Which of the following search algorithms is considered a depth-first search?

A) A* Search

B) Breadth-First Search

C) Iterative Deepening Search

D) Greedy Best-First Search

3

51. Which knowledge representation technique uses a graph of nodes and arcs?

A) Propositional Logic

B) Frame

C) Semantic Net

D) Production Rules

3

52. Which machine learning technique involves a teacher providing labeled data to the system?

A) Unsupervised Learning

B) Supervised Learning

C) Reinforcement Learning

D) Genetic Algorithm

2

53.Which neural network is known for its ability to minimize error using backpropagation?

A) Convolutional Neural Network (CNN)

B) Hopfield Network

C) Perceptron

D) Multilayer Perceptron (MLP)

4

54. In artificial neural networks, what is the role of an activation function?

A) To initialize the network

B) To normalize the input data

C) To introduce non-linearity into the network

D) To train the network

3

55.What type of projection shows an object as it would appear in 3D space using three dimensions?

A) Orthographic Projection

B) Pictorial View

C) Perspective Projection

D) Isometric Projection

4

56. Which of the following methods is used to calculate the Net Present Value (NPV) of a project?

A) Discounted Cash Flow

B) Payback Period

C) Internal Rate of Return (IRR)

D) Depreciation

1

57. What does a PERT chart help project managers to determine?

A) Critical Path

B) Budget

C) Staffing

D) Risk Management Plan

1

58.What does the term "scope creep" refer to in project management?

A) The expansion of project goals without corresponding adjustments to resources or time

B) The process of defining project goals

C) The unexpected increase in project budget

D) The gradual reduction of project scope

1

59. What is the role of engineers in societal development?

A) To only focus on technical solutions

B) To ensure their work aligns with societal needs and values

C) To minimize costs regardless of social impact

D) To prioritize profits over public welfare

2

60.Which document typically outlines the code of conduct for engineers?

A) Engineering Ethics Handbook

B) Code of Practice

C) Engineering Registration Act

D) National Engineering Policy

2

61.In a series circuit with a 12V battery connected to a resistor of 4Ω and a capacitor of 2μF, what is the time constant (τ) of the circuit?

A) 0.02 seconds

B) 0.008 seconds

C) 0.04 seconds

D) 0.1 seconds

1

62.A three-phase system has a line voltage of 400V. What is the phase voltage in a star (Y) connection?

A) 800V

B) 200.5V

C) 397.65V

D) 346.4V

4

63:What is the two's complement of -5 in an 8-bit signed number system?

A) 11111011

B) 11111010

C) 11111101

D) 10000011

1

64:If a JK flip-flop has J=1, K=1, and the previous state was 0, what will be the next state?

A) 0

B) 1

C) It will toggle

D) Undefined

3

65. What will be the output of the following C code?</p><p>#include <stdio.h>
void func(int *a) {
*a += 5;
}
int main() {
int x = 10;
func(&x);
printf("%d", x);
return 0;
}

A) 10

B) 15

C) 5

D) Compilation Error

2

66.What is the output of the following C++ program

#include <iostream>
using namespace std;</p><p>class Base {
public:
virtual void show() {
cout << "Base class" << endl;
}
};</p><p>class Derived : public Base {
public:
void show() override {
cout << "Derived class" << endl;
}
};</p><p>int main() {
Base *b; // Base class pointer
Derived d; // Derived class object
b = &d; // Assign derived class address to base class pointer
b->show(); // Call the show function
return 0;
}

A) Base class

B) Derived class

C) Compilation Error

D) Runtime Error

2

67. What is the size of the address bus in a microprocessor that can address 64KB of memory?

A) 16 bits

B) 18 bits

C) 32 bits

D) 64 bits

1

68.In an 8-bit microprocessor, how many different values can be represented?

A) 128

B) 255

C) 256

D) 512

3

69.In a network, if the data rate is 100 Mbps (megabits per second), how long will it take to transfer a file of size 1 GB (gigabyte)?

A) 80 seconds

B) 60 seconds

C) 100 seconds

D) 180 seconds

1

70.If a router processes a packet in 5 ms and the transmission delay is 10 ms for each packet over a link, what is the total time taken to transmit 20 packets?

A) 100 ms

B) 150 ms

C) 200 ms

D) 250 ms

3

71.Which of the following is true regarding the pumping lemma for context-free languages?

A) It is a property used to prove that certain languages are not context-free

B) It applies only to regular languages

C) It guarantees that all context-free languages are regular

D) It is only applicable to finite automata

1

72.A rectangle in a 2D space has vertices at (1, 2), (1, 4), (3, 4), and (3, 2). If the rectangle is scaled by a factor of 2 about the origin, what will be the coordinates of the vertex originally at (1, 2)?

A) (0.5, 1)

B) (1, 2)

C) (2, 4)

D) (2, 1)

3

73. What is the worst-case time complexity of accessing an element in a hash table?

A) O(n)

B) O(1)

C) O(log n)

D) O(n log n)

1

74.In a binary tree with 7 nodes, what is the maximum number of leaf nodes?

A) 3

B) 7

C) 5

D) 4

4

75.What does the acronym "CRUD" stand for in database management?

A) Create, Read, Update, Delete

B) Create, Review, Upload, Download

C) Connect, Retrieve, Update, Delete

D) Create, Run, Update, Destroy

1

76.In a software project, if the estimated number of lines of code (LOC) is 10,000 and the average productivity rate is 50 LOC per hour, how many hours will it take to complete the coding phase?

A) 150 hours

B) 200 hours

C) 250 hours

D) 300 hours

2

77. In reinforcement learning, what is the purpose of the exploration-exploitation trade-off?

A) To ensure the agent always chooses the action with the highest immediate reward.

B) To balance between exploring new actions and exploiting known rewarding actions.

C) To minimize the total number of actions taken during training.

D) To ensure that the agent does not learn from its environment.

2

78. If a neural network has 3 input neurons, 5 hidden neurons, and 2 output neurons, what is the total number of weights in the network?

A) 15

B) 20

C) 24

D) 30

3

79.If a project has an estimated duration of 12 weeks and requires 3 team members working full-time (40 hours per week), how many total man-hours are allocated for the project?

A) 1,440 hours

B) 1,200 hours

C) 960 hours

D) 480 hours

1

80.What is the main objective of risk management in project planning?

A) To avoid all risks at any cost.

B) To identify, assess, and prioritize risks to minimize their impact.

C) To eliminate uncertainties in project outcomes.

D) To ensure that all project tasks are completed on time.

2